

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

1st Semester of B. Pham Examination University Theory Examination April 2016 PH115 Pharmaceutical Engineering - I

Date: 14/04/2016 , Thursday Time: 10:00 a.m to 1:00 p.m Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
 2. **SECTION – I** comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to **SECTION – I** are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for **SECTION – I** is 30 minutes. Answers to **SECTION – I** must be written during the first 30 minutes of the examination.
 3. Answers to **SECTION – II** and **SECTION – III** are to be provided in separate Main Answer Books provided to you.
 4. Figures to right indicate marks.
 5. Draw neat sketches wherever necessary.
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SECTION - I

- Q 1** Attempt all questions. Each question is of one mark. **20**
1. Which one of the following is not a unit of pressure
[A] Bar
[B] Pascal
[C] mm of Hg
[D] gm/cm³
 2. From following what are the types of Glass?
[A] Borosilicate
[B] **sodalime**
[C] Treated soda lime
[D] All
 3. BTU is the unit of _____.
[A] Heat
[B] Volume
[C] Mass
[D] Temperature
 4. The mass of the component divided by the total mass of system is known as _____.
[A] Mole fraction
[B] Mass fraction
[C] % Mass
[D] Volume fraction
 5. Chemical or physical process, energy is neither created nor destroyed but one form of energy converted to another form.
[A] Dalton's Law
[B] Amagat's Law
[C] **1st Law of Thermodynamics**
[D] None

6. From the following which one is the graph type
- [A] Logarithmic
 - [B] Triangular
 - [C] A & B
 - [D] None
7. Sensible heat = _____ x special heat X temapature
- [A] Mass of water
 - [B] Size of vessel
 - [C] Pressure
 - [D] Vacuum
8. Rate of heat transfer = Driving force / _____
- [A] Temperatur
 - [B] Weight
 - [C] Resistance
 - [D] Volume
9. The amount of radiant energy emitted is proportion to the fourth power of the absolute temperature of the heat source body. – is the statement of _____ law.
- [A] Stefan Boltzamann
 - [B] Fourier's Law
 - [C] 2nd law of thermodinemics
 - [D] None
10. From following, which device is come in contact with direct heat
- [A] Rotary kiln
 - [B] Tunnel kiln
 - [C] Drum acid concentrator
 - [D] All
11. Which instrument is used for measure pressure of gas
- [A] Inclined manometer
 - [B] Rotameter
 - [C] Orifice Meter
 - [D] None
12. What are the way of lossed of energy in the pipe as per bernoulli's equation.
- [A] Friction losses
 - [B] End Effects
 - [C] Contraction Losses
 - [D] All
13. From the following which one is the non ferrous material used for paint construction.
- [A] Stainless Steel
 - [B] Mild Steel
 - [C] Tin
 - [D] None
14. What are the thermosettig plastic example?
- [A] Keebush
 - [B] LDP
 - [C] PVC
 - [D] Nylons

15. From the following which one is the name of valve.
 [A] Tapering Plug
 [B] Plug Clock
 [C] Swing Check
 [D] All
16. From the following which one is used for handling of suspension
 [A] Piston Pump
 [B] Plunger Pump
 [C] Diaphragm Pump
 [D] Gear Pump
17. Mass transfer takes place by bulk movement of the solution known as
 [A] Molecular Diffusion
 [B] Eddy Diffusion
 [C] Both
 [D] None
18. Which thing to be considered for design of mass transfer equipment?
 [A] Flow Condition
 [B] Concentration of material
 [C] Pressure
 [D] All
19. Pharmaceutical packaging material has the characteristic like.
 [A] Chemically Inert
 [B] Stable
 [C] Flexible
 [D] All
20. What are the disadvantages of glass as packaging material
 [A] Transparent
 [B] Flexible
 [C] Brittle
 [D] Heat Resistant

SECTION – II

- Q 2** Attempt any **FOUR** of the following
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|----------|--|-----------|
| A | Describe in detail different type of Graphical representation. | 05 |
| B | Explain in detail principle, working, advantage and disadvantage of Rotameter. | 05 |
| C | Derive Ideal gas law $PV = nRT$. | 05 |
| D | Describe steady state and unsteady state process with example. | 05 |
| E | Write a note on Prevention of Corrosion. | 05 |
| F | Short note on Diaphragm Valve. | 05 |

SECTION – III

- Q 3** Attempt any **FOUR** of the following
- A** Explain Heat transfer by Conduction. Derive equation for rate of heat transfer when resistance are in series 05
 - B** S.N. on Steam as heating medium. 05
 - C** Discuss advantage and limitation of different kind of Glass used in Pharma Industry 05
 - D** Describe various types of Packaging Material 05
 - E** Explain a) Tia Substance b) Dimensionless equation 05
 - F** Differentiate Orificemeter and Venturimeter 05
- Q 4** Attempt any **FOUR** of the following
- A** What is thermal Radiation? Explain concept of Black body. 05
 - B** Explain influence of Mass transfer on unit operation. 05
 - C** Explain Bernoulius Theorem 05
 - D** Factors affecting selection of material of pharmaceutical plant. 05
 - E** Explain Mechanism of type flow with Raynold's experiments. 05
 - F** Application of Heat Transfer in Industrial process. 05
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